



Semester One Examination, 2020

Question/Answer booklet

**MATHEMATICS
METHODS
UNIT 1**

**Section One:
Calculator-free**

SOLUTIONS

WA student number: In figures

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In words

Your name

Time allowed for this section

Reading time before commencing work:

five minutes

Working time:

fifty minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	8	8	50	52	35
Section Two: Calculator-assumed	13	13	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free

35% (52 Marks)

This section has **eight** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(5 marks)

The point $M(-2, 5)$ is the midpoint of point $A(-6, 3)$ and point B .

- (a) Determine the coordinates of point B .

(2 marks)

Solution
$\frac{-6 + x}{2} = -2 \Rightarrow x = 2$
$\frac{3 + y}{2} = 5 \Rightarrow y = 7$
$B(2, 7)$
Specific behaviours
✓ correct x -coordinate ✓ correct y -coordinate

- (b) Determine the equation of the straight line that passes through point $C(4, -1)$ and is perpendicular to the line through points A and M .

(3 marks)

Solution
$m_{AM} = \frac{5 - 3}{-2 - (-6)} = \frac{1}{2}$
$m_{\perp} = -1 \div \frac{1}{2} = -2$
$y - (-1) = -2(x - 4)$ $y = -2x + 7$
Specific behaviours
✓ gradient of AM ✓ perpendicular gradient ✓ correct equation

$m_{AM} = -\frac{1}{2}$ $m_{\perp} = 2$ $m_{AM} = -\frac{1}{4}$ $m_{\perp} = 4$
 $y = 2x - 9$ $y = 4x - 17$

Question 2

(4 marks)

Expand and simplify the following.

(a) $(x - 9)^2$.

(1 mark)

Solution
$(x - 9)^2 = x^2 - 18x + 81$
Specific behaviours
✓ correct expansion

(b) $(2x + 1)(x - 3)(x + 7)$.

(3 marks)

Solution
$\begin{aligned} (2x + 1)(x - 3)(x + 7) &= (2x + 1)(x^2 + 4x - 21) \\ &= 2x^3 + 8x^2 - 42x + x^2 + 4x - 21 \\ &= 2x^3 + 9x^2 - 38x - 21 \end{aligned}$
Specific behaviours
✓ correct binomial expansion ✓ correct full expansion ✓ simplifies correctly

$$(2x+1)(x-3)(x+7)$$

$$(2x^2+x-6x-3)(x+7)$$

$$(2x^2-5x-3)(x+7)$$

$$2x^3 - 5x^2 - 3x + 14x^2 - 35x - 21$$

$$2x^3 + 9x^2 - 38x - 21$$

Question 3

(5 marks)

Functions f and g are defined by $f(x) = 4x^2 - 4x + 5$ and $g(x) = 2x^2 - 8x + 6$.

- (a) Determine the discriminant of f and the discriminant of g .

(2 marks)

Solution
$\Delta_f = (-4)^2 - 4(4)(5) = -64$
$\Delta_g = (-8)^2 - 4(2)(6) = 16$
Specific behaviours
✓ discriminant of f
✓ discriminant of g

- (b) State, with justification, which function has no zeros and determine all zeros of the other function.

(3 marks)

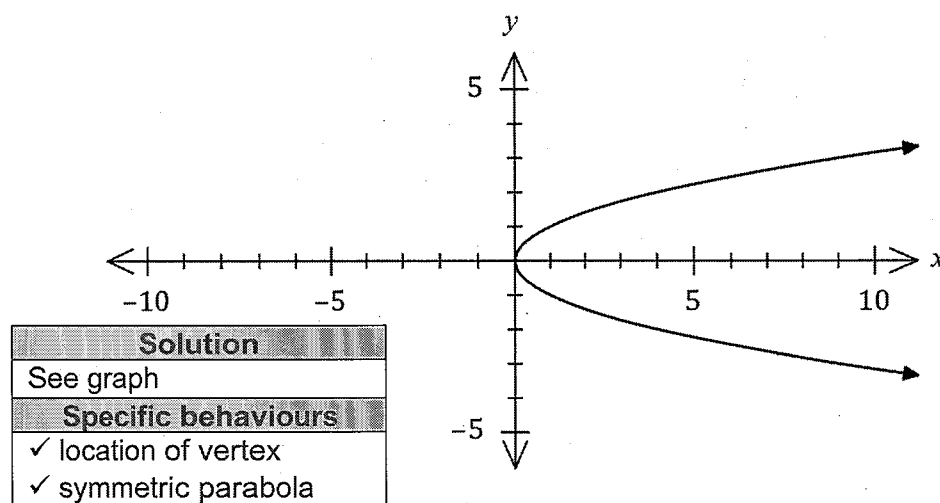
Solution
f has no zeroes as $\Delta_f < 0$.
$g(x) = 0$ when
$x = \frac{-(-8) \pm \sqrt{16}}{2(2)}$
$x = 2 \pm 1$
g has zeros when $x = 1, x = 3$.
Specific behaviours
✓ states f has no zeroes
✓ indicates appropriate method to find zeros
✓ both zeros of g

Question 4

(7 marks)

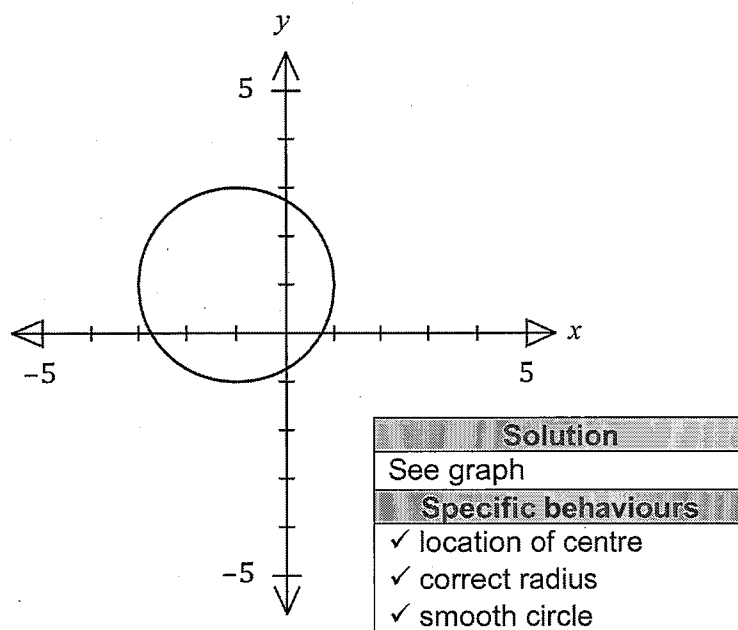
- (a) Sketch the graph of
- $y^2 = x$
- on the axes below.

(2 marks)



- (b) Sketch the graph of
- $(x + 1)^2 + (y - 1)^2 = 4$
- on the axes below.

(3 marks)



- (c) Explain whether
- y
- is a function of
- x
- in the relationship graphed in (a).

(2 marks)

Solution
y is NOT a function of x . This is because the graph of the relationship fails the vertical line test.
Specific behaviours
✓ states not a function
✓ explanation (VLT, one-to-many, etc)

Question 5

(8 marks)

(a) A periodic function is defined by $f(x) = 2 - 2 \sin(3x)$.

(i) State the amplitude of the function.

(1 mark)

Solution
Amplitude is 2.
Specific behaviours
✓ correct amplitude

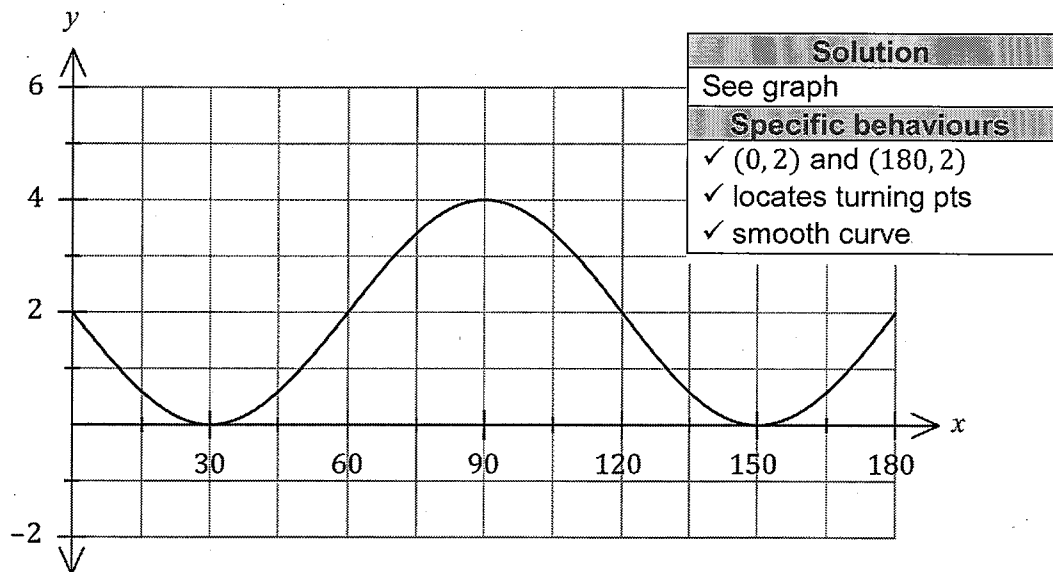
(ii) State the period of the function in degrees.

(1 mark)

Solution
Period is $360 \div 3 = 120^\circ$.
Specific behaviours
✓ correct period

(iii) Sketch the graph of $y = f(x)$ on the axes below.

(3 marks)



(b) Solve the equation $2 \cos(x - 15^\circ) = \sqrt{3}$ where $0 \leq x \leq 360^\circ$.

(3 marks)

Solution
$\cos(x - 15^\circ) = \frac{\sqrt{3}}{2}$ $x - 15^\circ = 30^\circ, 330^\circ$ $x = 45^\circ, 345^\circ$
Specific behaviours
✓ determines an angle for $\frac{\sqrt{3}}{2}$ ✓ determines one solution ✓ both correct solutions

Question 6

(7 marks)

(a) The variable V is inversely proportional to the variable t , so that when $t = 3.6$, $V = 10$.

(i) Explain how V will change as t increases.

(1 mark)

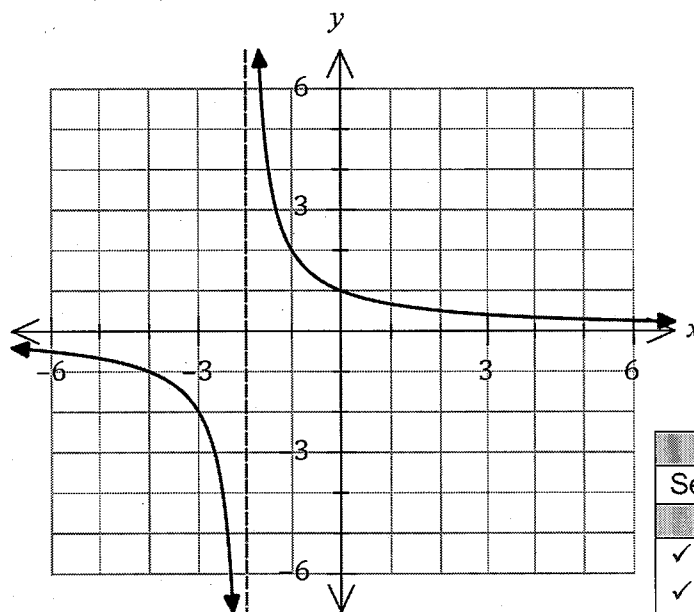
Solution
As t increases V will decrease.
Specific behaviours
✓ correct explanation

(ii) Determine t when $V = 3$.

(2 marks)

Solution
$V \times t = k \Rightarrow k = 3.6 \times 10 = 36$
$3t = 36 \Rightarrow t = 12$
Specific behaviours
✓ indicates appropriate method
✓ correct value

(b) Part of the graph of $y = \frac{a}{x+2}$ is drawn below.



Solution (b)(ii)
See graph
Specific behaviours
✓ asymptotes
✓ thru' (0, 1), (-4, -1)
✓ two smooth curves

(i) Determine the value of a .

(1 mark)

Solution
$(-1, 2) \Rightarrow 2 = \frac{a}{-1+2} \Rightarrow a = 2$
Specific behaviours
✓ correct value

(ii) Draw the remainder of the graph.

(3 marks)

Question 7

(8 marks)

Solve the following equations for x .

(a) $x^2 + 20x - 21 = 0$.

(2 marks)

Solution
$x^2 + 20x - 21 = (x - 1)(x + 21)$ Hence $x = 1, x = -21$.
Specific behaviours
✓ factorises ✓ states both solutions

(b) $(x - 1)^2 - 4 = 2x - 3$.

(3 marks)

Solution
$x^2 - 2x + 1 - 4 = 2x - 3$ $x^2 - 4x = 0$ $x(x - 4) = 0$ $x = 0, \quad x = 4$
Specific behaviours
✓ expands and equates to zero ✓ factorises ✓ states both solutions

(c) $x^3 - 2x^2 - 11x + 12 = 0$.

(3 marks)

Solution
$x^3 - 2x^2 - 11x + 12 _{x=1} = 13 - 13 = 0$ $x^3 - 2x^2 - 11x + 12 = (x - 1)(x^2 - x - 12)$ $= (x - 1)(x + 3)(x - 4)$ $x = -3, \quad x = 1, \quad x = 4$
Specific behaviours
✓ indicates linear factor ✓ factorises ✓ states all solutions

Question 8

(8 marks)

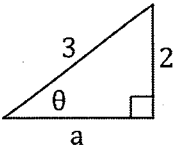
- (a) Determine an exact value for
- $\cos 103^\circ \cos 58^\circ + \sin 103^\circ \sin 58^\circ$
- .

(2 marks)

Solution
$\begin{aligned}\cos 103^\circ \cos 58^\circ + \sin 103^\circ \sin 58^\circ &= \cos(103^\circ - 58^\circ) \\ &= \cos 45^\circ \\ &= \frac{\sqrt{2}}{2}\end{aligned}$
Specific behaviours
✓ uses double angle formula ✓ states exact value

- (b) Determine all possible values of
- $\tan \theta$
- when
- $\sin \theta = \frac{2}{3}$
- .

(3 marks)

Solution
 $a = \sqrt{9 - 4} = \sqrt{5}$ Note that θ can be in quadrant 1 or 2. In Q1: $\tan \theta = \frac{2}{\sqrt{5}}$, in Q2 $\tan \theta = -\frac{2}{\sqrt{5}}$
Specific behaviours
✓ relevant use of right triangle ✓ one correct value ✓ both correct values

- (c) Determine an exact value for
- $\sin 75^\circ$
- .

(3 marks)

Solution
$\begin{aligned}\sin 75^\circ &= \sin(30^\circ + 45^\circ) \\ &= \sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ \\ &= \frac{1}{2} \times \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} \\ &= \frac{\sqrt{2} + \sqrt{6}}{4}\end{aligned}$
Specific behaviours
✓ uses double angle formula ✓ uses correct exact values ✓ simplifies to obtain final value

Supplementary page

Question number: _____



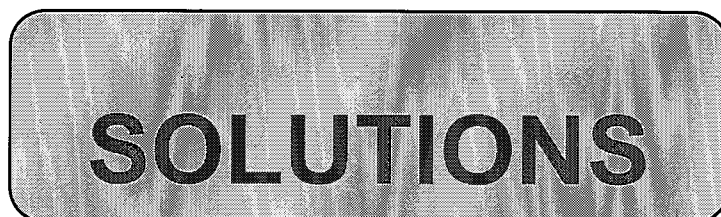
Semester One Examination, 2020

Question/Answer booklet

**MATHEMATICS
METHODS
UNIT 1**

Section Two:

Calculator-assumed



WA student number:

In figures

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In words

Your name

Time allowed for this section

Reading time before commencing work:

ten minutes

Working time:

one hundred

minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in this examination

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Section Two: Calculator-assumed

65% (98 Marks)

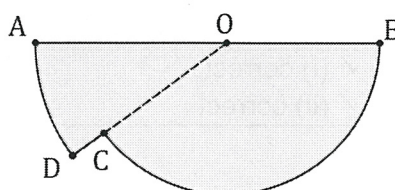
This section has **thirteen** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 9

(5 marks)

Shape $AOB CDA$ below consists of sector BOC of circle centre O joined to sector DOA of a different circle, also centre O . AB is a straight line of length 65 cm, arc AD is 12 cm long and $\angle AOD = 0.32$ radians.



- (a) Determine the length OA .

(2 marks)

Solution
$\text{Let } OA = R \text{ so that}$ $0.32R = 12$ $R = 37.5 \text{ cm}$
Specific behaviours
✓ correct use of arc length ✓ correct length

- (b) Determine the area of the shape.

(3 marks)

Solution
$A_{DOA} = \frac{1}{2} \times 37.5^2 \times 0.32$ $= 225$ ✓
$\text{Let } OB = r$ $r = 65 - 37.5$ $= 27.5$ ✓
$A_{BOC} = \frac{1}{2} \times 27.5^2 (\pi - 0.32)$ $= 1067$ ✓
$\text{Area} = 225 + 1067$ $= 1292 \text{ cm}^2$ ✓
Specific behaviours
✓ area of sector DOA ✓ radius and angle of sector BOC ✓ area of shape

$$(\pi - 0.32 = 2.822)$$

Question 10

(8 marks)

The height h metres of a particle above level ground is defined as a function of time t seconds as follows:

$$h(t) = 68.75 + 15t - 5t^2, \quad 0 \leq t \leq 5.5.$$

(a) Determine the height of the particle when

(i) $t = 0$.

(1 mark)

Solution
$h(0) = 68.75 \text{ m}$
$h(4.5) = 35 \text{ m}$
Specific behaviours
✓ (i) correct
✓ (ii) correct

(ii) $t = 4.5$.

(1 mark)

(b) Determine the maximum height reached by the particle and the time it reached this height.

(2 marks)

Solution
From graph of $h(t)$:
Maximum height: $h = 80 \text{ m}$ when $t = 1.5 \text{ s}$.
Specific behaviours
✓ correct height
✓ correct time

(c) Determine the time(s) that the particle was at a height of 75 m.

(2 marks)

Solution
From graph of $h(t)$:
$h = 75$ when $t = 0.5 \text{ s}, 2.5 \text{ s}$
Specific behaviours
✓ one time
✓ both times

(d) State the range of the function $h(t)$ for the given domain.

(2 marks)

Solution
Range of h :
$0 \leq h \leq 80$
Specific behaviours
✓ upper limit
✓ lower limit, correct inequality

Answer $0 \leq y \leq 80$

Question 11

(6 marks)

- (a) In city X, the fare for a 12.5 km taxi ride is \$18.30 and the fares for all taxi rides are directly proportional to the distance travelled.

Determine

- (i) the fare for a 25 km taxi ride.

(2 marks)

Solution
$\frac{25}{12.5} \times 18.30 = \36.60
Specific behaviours
✓ indicates use of direct proportion
✓ correct fare

- (ii) the distance a fare of \$25.62 would cover.

(2 marks)

Solution
$\frac{25.62}{18.30} \times 12.5 = 17.5 \text{ km}$
Specific behaviours
✓ indicates use of direct proportion
✓ correct distance

- (b) A straight line makes an angle of 60° with the positive x -axis and passes through the point with coordinates $(0, -6)$. Determine the exact equation of the line. (2 marks)

Solution
$m = \tan 60^\circ = \sqrt{3}$
$y = \sqrt{3}x - 6$
Specific behaviours
✓ gradient
✓ equation in exact form

and it is -6 see

Question 12

(8 marks)

The height above ground level, h m, of a seat on a steadily rotating Ferris wheel t minutes after the wheel begins to move is given by $h = 21.5 - 18.5 \cos\left(\frac{\pi t}{6} + \frac{\pi}{3}\right)$.

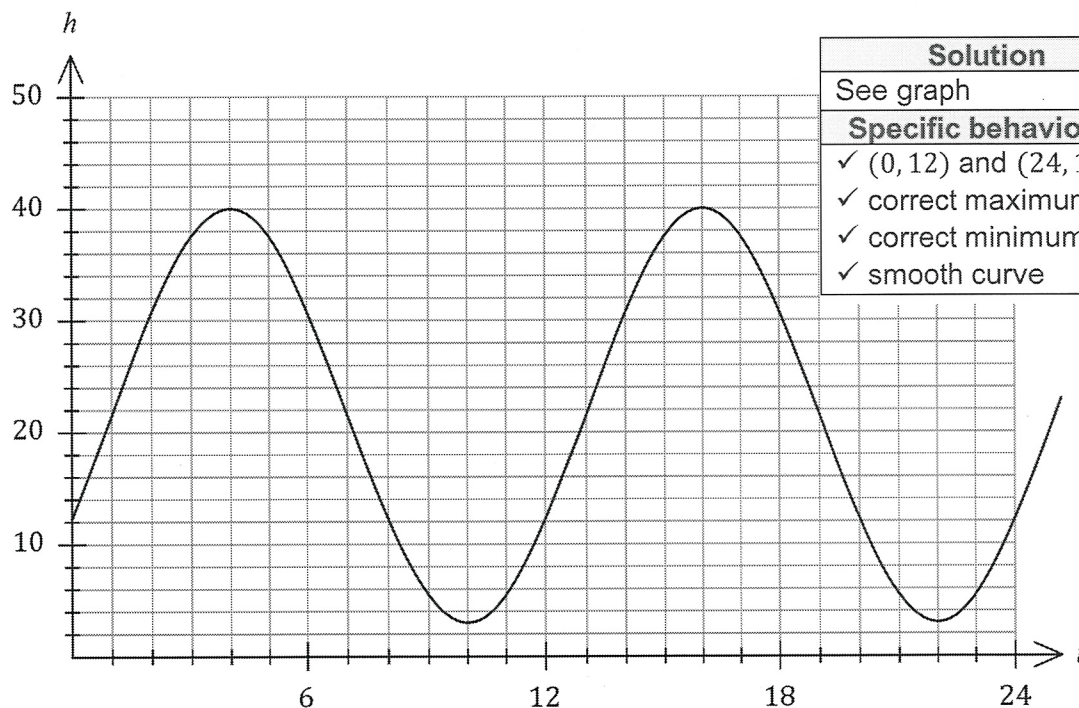
- (a) Determine the initial height of the seat.

(1 mark)

Solution
$h(0) = 12.25$ m
Specific behaviours
✓ correct height

- (b) Graph the height of the seat against time on the axes below.

(4 marks)



Solution
See graph
Specific behaviours
✓ (0, 12) and (24, 12)
✓ correct maximums
✓ correct minimums
✓ smooth curve

(4, 40), (16, 40)
(9, 2.25), (21, 2.25)

- (c) Determine

- (i) the maximum height above ground reached by the seat.

(1 mark)

Solution
$h_{MAX} = 40$ m
Specific behaviours
✓ correct height

- (ii) the time taken, to the nearest second, for the seat to first reach a height of 4 m above ground level.

(2 marks)

Solution
$h = 4 \Rightarrow t = 9.37$
$0.37 \times 60 = 22$
$t = 9 \text{ m } 22 \text{ s (562 s)}$
Specific behaviours
✓ time as decimal
✓ time to nearest second

Award ✓ if 9.37 seen

Question 13

(6 marks)

The graph $y = f(x)$, where $f(x) = x^2 + bx + c$ has a turning point at $(2, -7)$.

- (a) State the equation of the line of symmetry for the graph of $y = f(x)$.

(1 mark)

Solution
$x = 2$
Specific behaviours
✓ correct equation

- (b) Determine the value of the constant b and the value of the constant c .

(3 marks)

Solution
$f(x) = (x - 2)^2 - 7$ $= x^2 - 4x + 4 - 7$ $b = -4$ $c = -3$
Specific behaviours
✓ writes $f(x)$ in squared form ✓ value of b ✓ value of c

- (c) The graph of $y = f(x)$ is translated 3 units to the left and 2 units upwards. Determine the equation of the resulting curve.

(2 marks)

Solution
New turning point at $(2 - 3, -7 + 2) = (-1, -5)$.
Equation is $y = (x + 1)^2 - 5 = x^2 + 2x - 4$
Specific behaviours
✓ identifies new turning point ✓ correct equation (either form)

Question 14

(9 marks)

(a) Express

(i) 168° in radians.

(1 mark)

Solution
$168^\circ \times \frac{\pi}{180^\circ} = \frac{14\pi}{15}$
Specific behaviours
✓ correct measure

(ii) $\frac{5\pi}{12}$ in degrees.

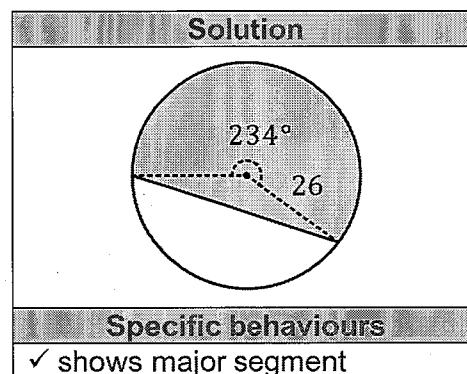
(1 mark)

Solution
$\frac{5\pi}{12} \times \frac{180^\circ}{\pi} = 75^\circ$
Specific behaviours
✓ correct measure

(b) A major segment subtends an angle of 234° in a circle of radius 26 cm.

(i) Sketch a diagram to show the circle and major segment.

(1 mark)



(ii) Determine the area of the major segment.

(3 marks)

Solution
$234^\circ \times \frac{\pi}{180^\circ} = \frac{13\pi}{10} = 4.084$
$A = \frac{1}{2}(26)^2 \left(\frac{13\pi}{10} - \sin \frac{13\pi}{10} \right)$
$A = 1\,654 \text{ cm}^2$
Specific behaviours
✓ uses radian measure ✓ substitutes correctly ✓ correct area

- (c) In triangle ABC , $AC = 65$ cm, $AB = 44$ cm and $\angle ACB = 37^\circ$. Determine the smallest possible area of the triangle. (3 marks)

Solution
$\frac{65}{\sin B} = \frac{44}{\sin 37^\circ}$
$B = 62.75^\circ \text{ or } 117.25^\circ$
For smallest area need $\angle A$ to be small as possible:
$A = 180^\circ - 37^\circ - 117.25^\circ = 25.75^\circ$
$\begin{aligned} \text{Area} &= \frac{1}{2} (65)(44) \sin 25.75^\circ \\ &= 621 \text{ cm}^2 \end{aligned}$
Specific behaviours
✓ size of angle B (Both given) ✓ smallest size of angle A ✓ correct area

Question 15**(8 marks)**

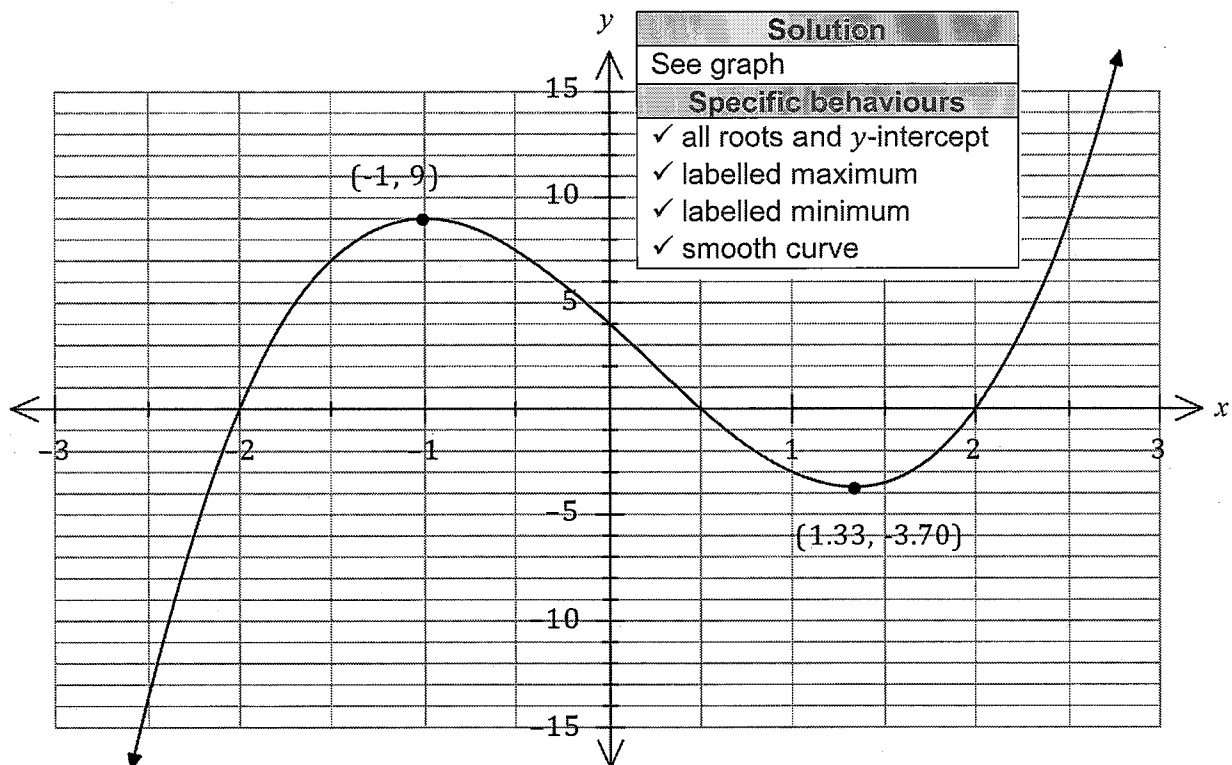
A polynomial of degree 3 passes through the points with coordinates $(0, 4)$, $(-2, 0)$, $(2, 0)$ and $(0.5, 0)$.

- (a) Determine the equation of the polynomial in expanded form.

(4 marks)

Solution
Using roots: $y = a(x + 2)(x - 2)(x - 0.5)$
Use 4th point: $x = 0 \Rightarrow 4 = a(2)(-2)(-0.5)$ $a = 2$
Expand: $y = 2(x + 2)(x - 2)(x - 0.5)$ $= 2x^3 - x^2 - 8x + 4$
Specific behaviours
✓ factored form using roots ✓ substitutes fourth point ✓ correct value of a ✓ correct expanded form

- (b) Draw the graph of the polynomial on the axes below, indicating the coordinates of all turning points.

(4 marks)

Question 16

(7 marks)

Let $f(x) = 1 + \sqrt{6 - 2x}$ and $g(x) = x + 2$.

(a) Evaluate $f(-5) - g(-5)$.

(2 marks)

Solution
$\begin{aligned} f(-5) - g(-5) &= 1 + \sqrt{16} - (-5 + 2) \\ &= 5 - (-3) \\ &= 8 \end{aligned}$
Specific behaviours
✓ evaluates f correctly ✓ correct value

(b) State the domain of $f(x)$.

(2 marks)

Solution
$\begin{aligned} 6 - 2x &\geq 0 \\ -2x &\geq -6 \\ x &\leq 3 \end{aligned}$
Specific behaviours
✓ correct inequality ✓ correct bound

(c) State the range of $g(x)$.

(1 mark)

Solution
$y \in \mathbb{R}$
Specific behaviours
✓ correct range (symbols or words)

(d) Determine the coordinates of the point(s) of intersection of $y = f(x)$ and $y = g(x)$.

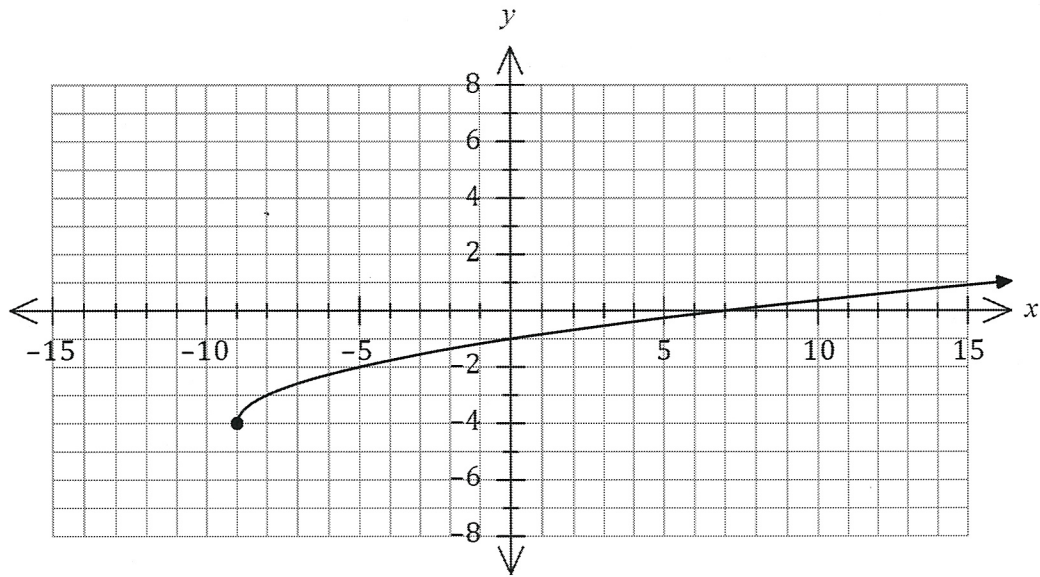
(2 marks)

Solution
Using graph/CAS: $(1, 3)$
Specific behaviours
✓ x-coordinate ✓ y-coordinate

Question 17

(8 marks)

The graph of $y = f(x)$ is drawn below, where $f(x) = \sqrt{x+a} + b$.



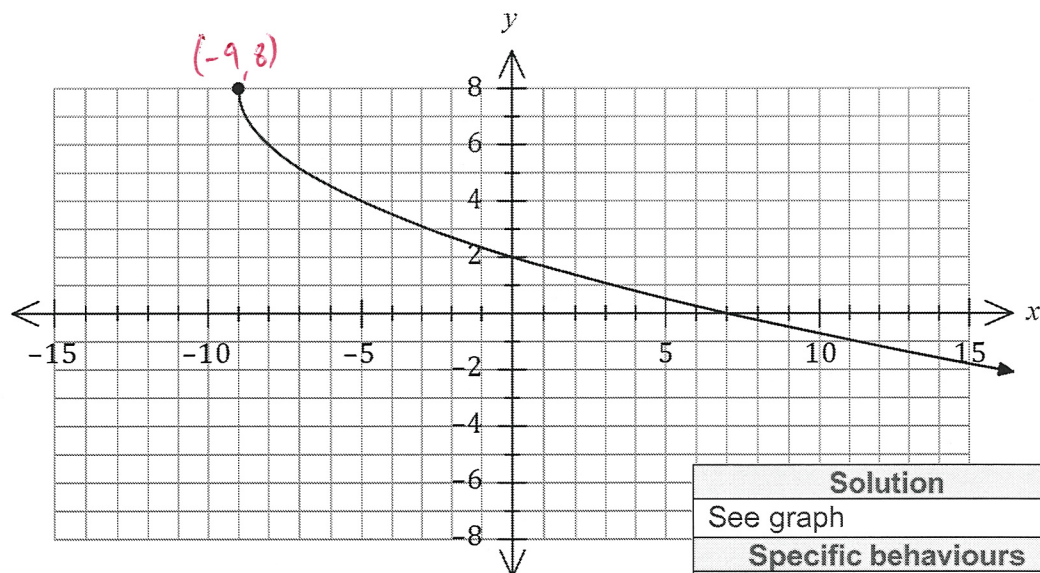
- (a) Determine the value of the constant a and the value of the constant b .

(2 marks)

Solution	
$a = 9,$	$b = -4$
Specific behaviours	
✓ value of a	
✓ value of b	

- (b) Draw the graph of $y = -2f(x)$ on the axes below.

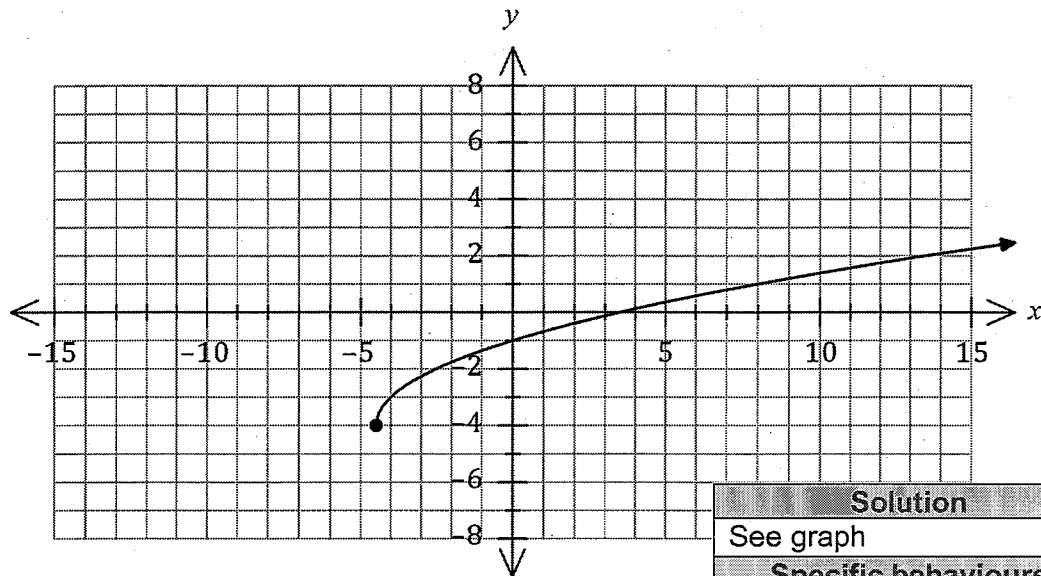
(3 marks)



Solution	
See graph	
Specific behaviours	
✓ endpoint at $(-9, 8)$	Start
✓ thru' $(0, 2)$ and $(7, 0)$	Intercepts ✓✓
✓ smooth curve	

(c) Draw the graph of $y = f(2x)$ on the axes below.

(3 marks)



Solution
See graph
Specific behaviours
✓ endpoint at $(-4.5, -4)$
✓ thru' $(0, -1)$ and $(3.5, 0)$
✓ smooth curve

Question 18

(5 marks)

The probabilities that Alf, Bess and Cleo will attend the next work social are $P(A) = 0.7$, $P(B) = 0.55$ and $P(C) = 0.85$ respectively. It is also known that $P(A \cap B) = 0.45$.

- (a) Determine $P(A \cup B)$.

(2 marks)

Solution
$P(A \cup B) = 0.7 + 0.55 - 0.45$ $= 0.8$
Specific behaviours
✓ uses probability rule ✓ correct probability

- (b) Describe, in the context of this question, the event $(A \cap \bar{B}) \cup (\bar{A} \cap B)$ and calculate the probability that it happens.

(3 marks)

Solution
The event means that either Alf or Bess but not both attend the next social. $P(A \cap \bar{B}) = 0.7 - 0.45 = 0.25$ $P(\bar{A} \cap B) = 0.55 - 0.45 = 0.1$ $P = 0.25 + 0.1 = 0.35$
Specific behaviours
✓ description ✓ one correct part probability ✓ correct answer

Question 14 19

(6 marks)

When a random sample of 173 people from a university were classified according to whether they had a driver's licence (event D) and whether they wore spectacles (event S), it was observed that $n(D) = 140$, $n(S) = 53$ and $n(S \cap D) = 10$.

(a) Determine

(i) $n(\bar{S})$.

(1 mark)

Solution
$n(\bar{S}) = 173 - 53 = 120$
Specific behaviours
✓ correct number

(ii) $n(D \cap S)$.

(1 mark)

Solution
$n(D \cap S) = 53 - 10 = 43$
Specific behaviours
✓ correct number

(b) Determine the probability that a randomly chosen person from the sample

(i) does not have a driver's licence.

(2 marks)

Solution
$n(\bar{D}) = 173 - 140 = 33$
$P(\bar{D}) = \frac{33}{173} \approx 0.191$
Specific behaviours
✓ numerator
✓ denominator

(ii) wears spectacles given that they have a driver's licence.

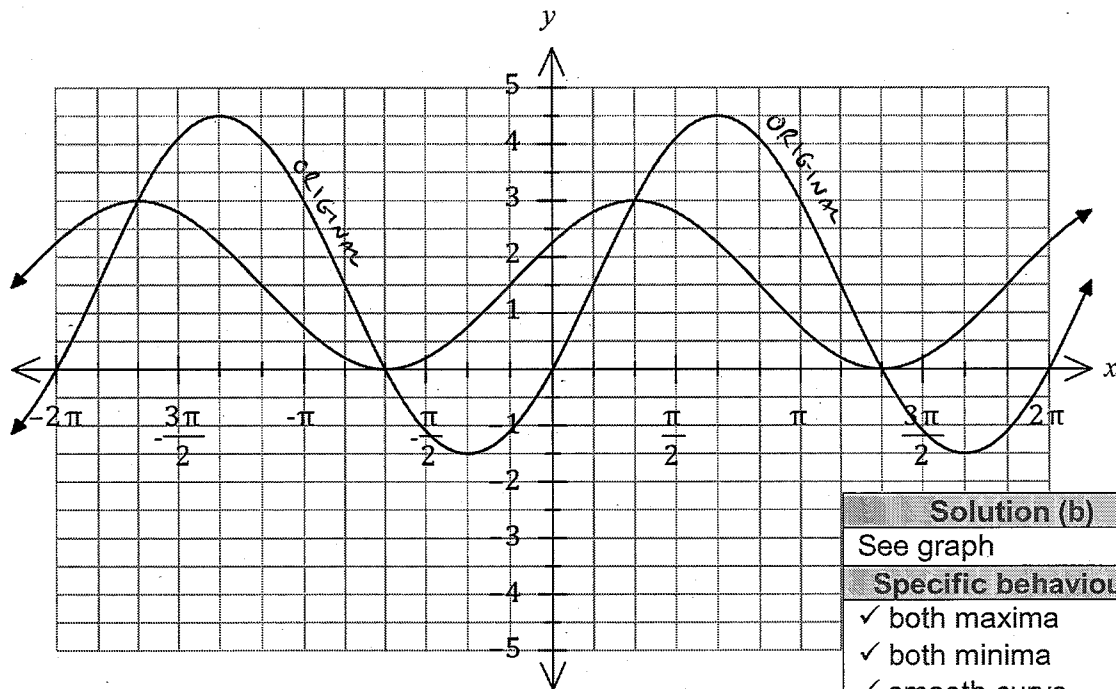
(2 marks)

Solution
$P(S D) = \frac{n(D \cap S)}{n(D)} = \frac{43}{140} \approx 0.307$
Specific behaviours
✓ numerator
✓ denominator

Question 20

(8 marks)

The graph of $y = a + b \sin(x - c)$ is drawn below, where a, b and c are positive constants.

**Solution (b)**

See graph

Specific behaviours

- ✓ both maxima
- ✓ both minima
- ✓ smooth curve

- (a) Determine the value of a , the value of b and the value of c , where $c < \pi$. (3 marks)

Solution

$$a = 1.5, \quad b = 3, \quad c = \frac{\pi}{6}$$

Specific behaviours

- ✓ value of a
- ✓ value of b
- ✓ value of c

- (b) On the same axes, draw the graph of $y = a + \frac{b}{2} \sin(x + c)$. (3 marks)

- (c) Solve $b \sin(x - c) = \frac{b}{2} \sin(x + c)$ for $-\pi \leq x \leq \pi$. (2 marks)

Solution

Using intersection of graphs:

$$x = -\frac{2\pi}{3}, \quad x = \frac{\pi}{3}$$

Specific behaviours

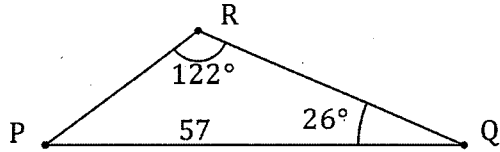
- ✓ a correct solution, anywhere
- ✓ two solutions as given

Question 21

(8 marks)

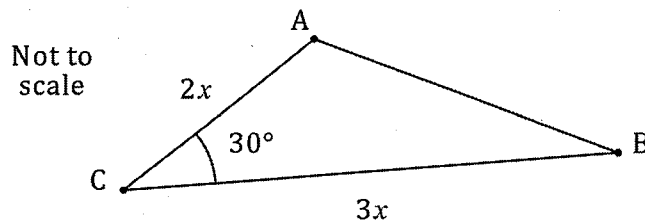
- (a) Determine the area of triangle PQR when $\angle PQR = 26^\circ$, $\angle PRQ = 122^\circ$ and $PQ = 57$ cm.

(4 marks)

Solution
 $\frac{q}{\sin 26^\circ} = \frac{57}{\sin 122^\circ} \Rightarrow q = 29.46$ $\text{Area} = \frac{1}{2}(57)(29.46) \sin(32^\circ)$ $= 445 \text{ cm}^2$
Specific behaviours
✓ sketch of triangle ✓ correct use of sine rule ✓ length of second side ✓ correct area

- (b) The area of triangle ABC is 96 cm^2 , $\angle ACB = 30^\circ$ and $2BC = 3AC$ as shown in the diagram. Determine the length of AB .

(4 marks)

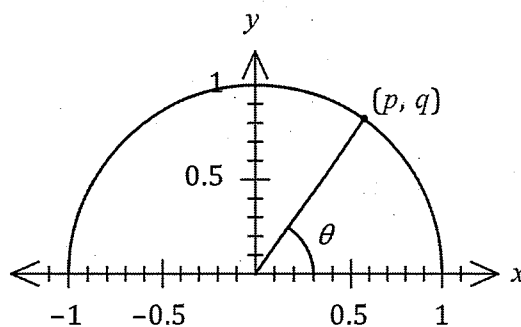


Solution
$\frac{1}{2}(2x)(3x) \sin(30^\circ) = 96$ $x = 8$ $AB^2 = 16^2 + 24^2 - 2(16)(24) \cos(30^\circ)$ $AB = 12.92 \text{ cm}$
Specific behaviours
✓ area equation ✓ value of x ✓ cosine rule ✓ length of AB

Question 22

(6 marks)

Consider part of the unit circle shown below, where $\theta = 51^\circ$.



Determine, in terms of p and / or q , an expression for each of the following:

(a) $\sin 51^\circ$.

(1 mark)

Solution
$\sin 51^\circ = q$
Specific behaviours
✓ correct expression

(b) $\sin 141^\circ$.

(2 marks)

Solution
$\sin 141^\circ = \sin(51^\circ + 90^\circ)$ $= \cos 51^\circ$ $= p$
Specific behaviours
✓ uses fundamental relationship ✓ correct expression

(c)
(d) $\cos 258^\circ$.

(3 marks)

Solution
$\cos 258^\circ = \cos(129^\circ + 129^\circ)$ $= \cos 129^\circ \cos 129^\circ - \sin 129^\circ \sin 129^\circ$ $= (-p)^2 - (q)^2$ $= p^2 - q^2$
Specific behaviours
✓ halves angle ✓ uses sum identity ✓ correct expression